ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION **WINTER SEMESTER, 2023-2024**

DURATION: 3 HOURS **FULL MARKS: 150**

Math 4341: Linear Algebra

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 6 (six)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

---

1. a) Given, 15 (CO3) (PO1)

$$A = \begin{bmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{bmatrix}$$

Find the Singular Value Decomposition (SVD) of the matrix A and express it in the form $A = U\Sigma V^T$, identifying the matrices U, Σ and V explicitly.

Solution:

Given,

$$A = \begin{bmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{bmatrix}$$

The transpose of A (A^T) is:

$$A^T = \begin{bmatrix} -3 & 6 & 6 \\ 1 & -2 & -2 \end{bmatrix}$$

Now, $A^T A$:

$$A^T A = \begin{bmatrix} -3 & 6 & 6 \\ 1 & -2 & -2 \end{bmatrix} \begin{bmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{bmatrix} = \begin{bmatrix} 81 & -27 \\ -27 & 9 \end{bmatrix}$$

To find the eigenvalues, we solve the characteristic equation:

$$\det(A^T A - \lambda I) = 0$$

Where λ represents the eigenvalues and I is the identity matrix.

$$A^T A - \lambda I = \begin{bmatrix} 81 - \lambda & -27 \\ -27 & 9 - \lambda \end{bmatrix}$$

The determinant is:

$$\det(A^T A - \lambda I) = (81 - \lambda)(9 - \lambda) - (-27)(-27) = 0$$

Expanding:

$$729 - 81\lambda - 9\lambda + \lambda^2 - 729 = 0$$

Simplifying:

$$\lambda^2 - 90\lambda = 0$$

Factoring:

$$\lambda(\lambda - 90) = 0$$

Thus, the eigenvalues are:

$$\lambda_1 = 0, \quad \lambda_2 = 90$$

Now, finding eigenvectors of $A^T A$:
For $\lambda = 90$ (descending order) :

$$\begin{bmatrix} 81 & -27 \\ -27 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} 90x_1 \\ 90y_1 \end{bmatrix}$$

Setting, $y_1 = 1$ we get,

$$81x_1 - 27(y_1) = 90x_1$$
$$\Rightarrow 9x_1 = -27$$
$$\Rightarrow x_1 = -3$$
$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

After normalizing,

$$\therefore v_1 = \begin{bmatrix} \frac{-3}{\sqrt{10}} \\ \frac{-3}{\sqrt{10}} \end{bmatrix}$$

For $\lambda = 0$ (descending order) :

$$\begin{bmatrix} 81 & -27 \\ -27 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Setting, $x_1 = 1$ we get,

$$81x_1 - 27(y_1) = 0$$
$$\Rightarrow 27y_1 = 81(1)$$
$$\Rightarrow y_1 = 3$$
$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

After normalizing,

$$\therefore v_2 = \begin{bmatrix} \frac{1}{\sqrt{10}} \\ \frac{3}{\sqrt{10}} \end{bmatrix}$$

Thus,

$$V = \begin{bmatrix} \frac{-3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \\ \frac{-3}{\sqrt{10}} & \frac{3}{\sqrt{10}} \end{bmatrix}$$

Again,

$$AA^T = \begin{bmatrix} 10 & -20 & -20 \\ -20 & 40 & 40 \\ -20 & 40 & 40 \end{bmatrix}$$

,

For $\lambda = 90$,

$$\begin{bmatrix} 10 & -20 & -20 \\ -20 & 40 & 40 \\ -20 & 40 & 40 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 90x_1 \\ 90y_1 \\ 90z_1 \end{bmatrix}$$

We consider $y_1 = -2$ and $z_1 = -2$, so,

$$10x_1 - 20(-2) - 20(-2) = 90$$
$$\Rightarrow x_1 = 1$$

$$\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ -2 \end{bmatrix} =$$

Normalizing,

$$u_1 = \begin{bmatrix} \frac{1}{3} \\ \frac{-2}{3} \\ \frac{-2}{3} \end{bmatrix}$$

Similarly, for $\lambda = 0$,
we set $y_1 = 1$ and $z_1 = 0$ once, and get

$$u_2 = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \\ 0 \end{bmatrix}$$

and, we set $y_1 = 0$ and $z_1 = 1$, and get

$$u_3 = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ 0 \\ \frac{1}{\sqrt{5}} \end{bmatrix}$$

$$\therefore U = \begin{bmatrix} \frac{1}{3} & \frac{2}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ \frac{-2}{3} & \frac{1}{\sqrt{5}} & 0 \\ \frac{-2}{3} & 0 & \frac{1}{\sqrt{5}} \end{bmatrix}$$

Thus,

$$A = U\Sigma V^T$$

$$\Rightarrow A = \begin{bmatrix} \frac{1}{3} & \frac{2}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ \frac{-2}{3} & \frac{1}{\sqrt{5}} & 0 \\ \frac{-2}{3} & 0 & \frac{1}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} \sqrt{90} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{-3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \\ \frac{-3}{\sqrt{10}} & \frac{3}{\sqrt{10}} \end{bmatrix}^T$$

V and U might be different in your solution

Rubric:

- 1 point for correctly writing $A^T A$ and AA^T.
- 1 point each for correctly identifying eigenvalues (times 2).
- 2 points for each for correctly identifying each eigenvector (times 5).

- 1 point for correctly identifying Σ.
- 1 point for correct $A = U\Sigma V^T$ representation.

b) Pell numbers form a unique sequence that appears in various mathematical contexts, from ancient problems to modern applications. They are defined by a recurrence relation and follow a specific pattern. The sequence of Pell numbers are as follows:

$$0, 1, 2, 5, 12, 29, 70, 169, \cdots$$

Each Pell number is derived from a specific combination of the two prior numbers.

i. Find the equation to determine the n-th Pell number, P_n. 2 (CO3) (PO1)

Solution:

The sequence of Pell numbers is given as: 0, 1, 2, 5, 12, 29, 70, 169, ...
Let's look at the relationship between a term and the two preceding terms.
Consider the general linear homogeneous recurrence relation of order 2:

$$P_n = aP_{n-1} + bP_{n-2}$$

Using the first few terms of the sequence:
For n = 2: $P_2 = 2 = aP_1 + bP_0 = a(1) + b(0) \Rightarrow a = 2$
For n = 3: $P_3 = 5 = aP_2 + bP_1 = 2(2) + b(1) = 4 + b \Rightarrow b = 1$
Let's verify this recurrence relation with the next term: For n = 4: $P_4 = 2P_3 + 1P_2 = 2(5) + 1(2) = 10 + 2 = 12$. This matches the sequence.
Therefore, the equation to determine the n-th Pell number, P_n, is given by the recurrence relation:

$$\boxed{P_n = 2P_{n-1} + P_{n-2}}$$

with initial conditions $P_0 = 0$ and $P_1 = 1$.

Rubric:

- 1 point each for correctly determining the values of a and b.

ii. Using the equation obtained in Question 1.b)i, express the n-th Pell number using matrix diagonalization. 7 (CO3) (PO1)

Solution:

We can express the recurrence relation $P_n = 2P_{n-1} + P_{n-2}$ in matrix form as:

$$\begin{bmatrix} P_n \\ P_{n-1} \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} P_{n-1} \\ P_{n-2} \end{bmatrix}$$

Let $M = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}$ and $V_n = \begin{bmatrix} P_n \\ P_{n-1} \end{bmatrix}$. Then, $V_n = MV_{n-1}$. By repeated application, we get $V_n = M^n V_0$, where $V_0 = \begin{bmatrix} S_1 \\ S_0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.
To find M^n, we can use matrix diagonalization. First, we find the eigenvalues of M by solving the characteristic equation $\det(M - \lambda I) = 0$:

$$\det \begin{bmatrix} 2-\lambda & 1 \\ 1 & -\lambda \end{bmatrix} = (2-\lambda)(-\lambda) - (1)(1) = -2\lambda + \lambda^2 - 1 = \lambda^2 - 2\lambda - 1 = 0$$

Using the quadratic formula, the eigenvalues are:

$$\lambda = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(-1)}}{2(1)} = \frac{2 \pm \sqrt{4+4}}{2} = \frac{2 \pm \sqrt{8}}{2} = \frac{2 \pm 2\sqrt{2}}{2} = 1 \pm \sqrt{2}$$

Let $\lambda_1 = 1 + \sqrt{2}$ and $\lambda_2 = 1 - \sqrt{2}$.
Next, we find the eigenvectors corresponding to each eigenvalue.
For $\lambda_1 = 1 + \sqrt{2}$:

$$(M - \lambda_1 I)v_1 = \begin{bmatrix} 2 - (1+\sqrt{2}) & 1 \\ 1 & -(1+\sqrt{2}) \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow (1 - \sqrt{2})x + y = 0$$

$$\Rightarrow \ y = (\sqrt{2} - 1)x$$

Taking $x = 1, y = (\sqrt{2} - 1)$ or $-(1 - \sqrt{2})$ or $-\lambda_2$.
$\therefore$The eigenvector is $v_1 = \begin{bmatrix} 1 \\ \sqrt{2} - 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -\lambda_2 \end{bmatrix}$.
For $\lambda_2 = 1 - \sqrt{2}$:

$$(M - \lambda_2 I)v_2 = \begin{bmatrix} 2 - (1-\sqrt{2}) & 1 \\ 1 & -(1-\sqrt{2}) \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$(1 + \sqrt{2})x + y = 0 \ \Rightarrow \ y = -(1 + \sqrt{2})x$$

Taking $x = 1$,$y = (-\sqrt{2} - 1)$ or $-(1 + \sqrt{2})$ or $-\lambda_1$.
$\therefore$The eigenvector is $v_2 = \begin{bmatrix} 1 \\ -1 - \sqrt{2} \end{bmatrix} = \begin{bmatrix} 1 \\ -\lambda_1 \end{bmatrix}$.
Now, we form the matrix S with the eigenvectors as columns and the diagonal matrix Λ with the eigenvalues on the diagonal:

$$S = \begin{bmatrix} 1 & 1 \\ -\lambda_2 & -\lambda_1 \end{bmatrix}, \quad D = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$

We have $M = S\Lambda S^{-1}$. Then $M^n = S\Lambda^n S^{-1}$.

$$\Lambda^n = \begin{bmatrix} \lambda_1^n & 0 \\ 0 & \lambda_2^n \end{bmatrix}$$

The inverse of S is $S^{-1} = \frac{1}{\det(S)} \begin{bmatrix} -\lambda_1 & -1 \\ \lambda_2 & 1 \end{bmatrix}$.

$$\det(P) = -\lambda_1 + \lambda_2 = \lambda_2 - \lambda_1$$

$$P^{-1} = \frac{1}{\lambda_2 - \lambda_1} \begin{bmatrix} -\lambda_1 & -1 \\ \lambda_2 & 1 \end{bmatrix}$$

Now, $V_n = M^n V_1 = S\Lambda^n S^{-1} V_0$:

$$\Rightarrow \begin{bmatrix} S_n \\ S_{n-1} \end{bmatrix} = S\Lambda^n S^{-1} \begin{bmatrix} P_1 \\ P_0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} P_n \\ P_{n-1} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -\lambda_1 & -\lambda_1 \end{bmatrix} \begin{bmatrix} \lambda_1^n & 0 \\ 0 & \lambda_2^n \end{bmatrix} \frac{1}{\lambda_2 - \lambda_1} \begin{bmatrix} -\lambda_1 & -1 \\ \lambda_2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} P_n \\ P_{n-1} \end{bmatrix} = \frac{1}{\lambda_2 - \lambda_1} \begin{bmatrix} 1 & 1 \\ -\lambda_1 & -\lambda_1 \end{bmatrix} \begin{bmatrix} \lambda_1^n & 0 \\ 0 & \lambda_2^n \end{bmatrix} \begin{bmatrix} -\lambda_1 \\ \lambda_2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} P_n \\ P_{n-1} \end{bmatrix} = \frac{1}{\lambda_2 - \lambda_1} \begin{bmatrix} 1 & 1 \\ -\lambda_1 & -\lambda_1 \end{bmatrix} \begin{bmatrix} -\lambda_1^{n+1} \\ \lambda_2^{n+1} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} P_n \\ P_{n-1} \end{bmatrix} = \frac{1}{\lambda_2 - \lambda_1} \begin{bmatrix} -\lambda_1^{n+1} + \lambda_2^{n+1} \\ \lambda_2\lambda_1^{n+1} - \lambda_1\lambda_2^{n+1} \end{bmatrix}$$

The n-th Pell number S_n is the first component of this vector:

$$S_n = \frac{1}{\lambda_2 - \lambda_1}[\lambda_2^{n+1} - \lambda_1^{n+1}]$$

$$\boxed{P_n = \frac{\lambda_2^{n+1} - \lambda_1^{n+1}}{\lambda_2 - \lambda_1} = \frac{(1-\sqrt{2})^{n+1} - (1+\sqrt{2})^{n+1}}{-2\sqrt{2}}}$$

Rubric:

- 1 point for correctly determining eigenvalues.
- 2 points each for correctly determining each eigenvector (times 2).
- 2 points for correctly diagonalizing and finding out the n-th term.

iii. Show that the 13th Pell number is $33,461$ using the result obtained in Question 1.b)ii. 1 (CO1) (PO1)

Solution:

From Question 1.b)ii, we got,

$$\boxed{P_n = \frac{\lambda_2^{n+1} - \lambda_1^{n+1}}{\lambda_2 - \lambda_1} = \frac{(1-\sqrt{2})^{n+1} - (1+\sqrt{2})^{n+1}}{-2\sqrt{2}}}$$

The 13th term (0-indexed series), $n = 12$,

$$P_n = \frac{(1-\sqrt{2})^{12+1} - (1+\sqrt{2})^{12+1}}{-2\sqrt{2}} = 33461$$

Rubric:

- 0.5 point for correctly determining n value.
- 0.5 point for right answer.

2. Given,

$$\vec{v}_1 = \begin{bmatrix} 2 \\ 0 \\ 2 \\ -1 \\ 0 \end{bmatrix}, \quad \vec{v}_2 = \begin{bmatrix} 0 \\ 3 \\ 0 \\ 0 \\ -4 \end{bmatrix}, \quad \vec{v}_3 = \begin{bmatrix} 1 \\ 1 \\ 2 \\ -3 \\ 2 \end{bmatrix}, \quad \vec{v}_4 = \begin{bmatrix} -1 \\ 2 \\ 1 \\ 3 \\ 1 \end{bmatrix} \quad \text{and} \quad R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

a) For any invertible matrix B, show that 4 (CO3) (PO1)

$$B^{-1}\vec{x} = \frac{1}{\lambda}\vec{x}$$

where λ and $\vec{x}$ denote the eigenvalue and the eigenvector of the matrix B respectively.

Solution:

We are given that $B\vec{x} = \lambda\vec{x}$, where λ is the eigenvalue and $\vec{x}$ is the eigenvector of B.

$$\Rightarrow B\vec{x} = \lambda\vec{x}$$

Multiplying both sides by B^{-1}:

$$\Rightarrow B^{-1}B\vec{x} = B^{-1}\lambda\vec{x}$$

Since $B^{-1}B = I$, we get:

$$\Rightarrow \vec{x} = \lambda B^{-1}\vec{x}$$

Dividing both sides by λ (assuming $\lambda \neq 0$):

$$\Rightarrow B^{-1}\vec{x} = \frac{1}{\lambda}\vec{x}$$

Thus, we have shown that:

$$B^{-1}\vec{x} = \frac{1}{\lambda}\vec{x}$$

Rubric:

- 1 point for starting with $B\vec{x} = \lambda\vec{x}$.
- 1 point for multiplying the expression with B^{-1}.
- 1 point for multiplying the expression with $\frac{1}{\lambda}$.
- 1 point for correctly showing $B^{-1}\vec{x} = \frac{1}{\lambda}\vec{x}$.

b) Determine the eigenvalues and eigenvectors of the matrix R for $\theta = 90^\circ$ and $\theta = 180^\circ$ separately, and logically justify your answer for each case. 5 + 5 (CO3) (PO1)

Solution:

To determine the eigenvalues and eigenvectors of the rotation matrix R for specific values of θ, we'll calculate them separately for $\theta = 90°$ and $\theta = 180°$. The general form of the rotation matrix is:

$$R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Case 1: $\theta = 90°$ Substituting $\theta = 90°$ into the rotation matrix:

$$R = \begin{bmatrix} \cos 90° & -\sin 90° \\ \sin 90° & \cos 90° \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

We want to find the eigenvalues and eigenvectors of the matrix R. To do this, we need to solve the characteristic equation:

$$\det(R - \lambda I) = 0$$

Where I is the identity matrix, and λ is the eigenvalue.

$$R - \lambda I = \begin{bmatrix} 0-\lambda & -1 \\ 1 & 0-\lambda \end{bmatrix} = \begin{bmatrix} -\lambda & -1 \\ 1 & -\lambda \end{bmatrix}$$

Now, calculate the determinant:

$$\det(R - \lambda I) = (-\lambda)(-\lambda) - (-1)(1) = \lambda^2 + 1$$

Setting the determinant equal to zero to solve for λ:

$$\lambda^2 + 1 = 0$$

$$\lambda^2 = -1$$

$$\lambda = \pm i$$

So the eigenvalues are $\lambda_1 = i$ and $\lambda_2 = -i$.

Eigenvalue $\lambda = i$: To find the corresponding eigenvector for $\lambda = i$, substitute $\lambda = i$ into the equation $(R - \lambda I)\vec{x} = 0$:

$$\begin{bmatrix} -i & -1 \\ 1 & -i \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This leads to the system of equations:

$$-ix_1 - x_2 = 0 \Rightarrow x_2 = -ix_1$$

$$x_1 - ix_2 = 0 \Rightarrow x_1 - i(-ix_1) = x_1 - x_1 = 0$$

Let $x_1 = 1$, then $x_2 = -i$. The eigenvector corresponding to $\lambda = i$ is $\vec{x} = \begin{bmatrix} 1 \\ -i \end{bmatrix}$.

Eigenvalue $\lambda = -i$: To find the corresponding eigenvector for $\lambda = -i$, substitute $\lambda = -i$ into the equation $(R - \lambda I)\vec{x} = 0$:

$$\begin{bmatrix} i & -1 \\ 1 & i \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This leads to the system of equations:

$$ix_1 - x_2 = 0 \Rightarrow x_2 = ix_1$$

$$x_1 + ix_2 = 0 \Rightarrow x_1 + i(ix_1) = x_1 - x_1 = 0$$

Let $x_1 = 1$, then $x_2 = i$. The eigenvector corresponding to $\lambda = -i$ is $\vec{x} = \begin{bmatrix} 1 \\ i \end{bmatrix}$.

Logical Justification for $\theta = 90°$: A rotation by 90° in the counter-clockwise direction transforms a real vector in the 2D plane. No non-zero real vector will simply be scaled by this transformation; instead, its direction changes. This suggests that the eigenvalues cannot be real (unless they are 1 or -1, which is not the case here). The complex eigenvalues i and $-i$ indicate that the eigenvectors must involve complex components, which is consistent with the mathematical derivation. The eigenvectors we found have complex entries, reflecting the fact that the scaling by a purely imaginary factor inherently involves a change in the vector's direction relative to the real axes.

Case 2: $\theta = 180°$ Substituting $\theta = 180°$ into the rotation matrix:

$$R = \begin{bmatrix} \cos 180° & -\sin 180° \\ \sin 180° & \cos 180° \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

Again, we solve for the eigenvalues by finding the determinant of $R - \lambda I$:

$$R - \lambda I = \begin{bmatrix} -1-\lambda & 0 \\ 0 & -1-\lambda \end{bmatrix}$$

The determinant is:

$$\det(R - \lambda I) = (-1-\lambda)(-1-\lambda) - (0)(0) = (\lambda+1)^2$$

Setting the determinant equal to zero:

$$(\lambda+1)^2 = 0$$

Thus, $\lambda = -1$ is the only eigenvalue, with algebraic multiplicity 2.

Eigenvector for $\lambda = -1$: To find the eigenvector corresponding to $\lambda = -1$, substitute $\lambda = -1$ into the equation $(R - \lambda I)\vec{x} = 0$:

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This system yields no additional restrictions on x_1 and x_2, meaning the eigenvector can be any non-zero vector of the form $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. Thus, any non-zero vector in the 2D plane is an eigenvector for $\lambda = -1$. We can choose two linearly independent eigenvectors, for example:

$$\vec{x}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \vec{x}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Logical Justification for $\theta = 180°$: A rotation by 180° in the counter-clockwise direction reverses the direction of any vector in the 2D plane. This means that the transformed vector is -1 times the original vector. Therefore, every non-zero vector is an eigenvector with the eigenvalue -1. This is consistent with the mathematical derivation where we found a repeated eigenvalue of -1 and showed that any vector satisfies the eigenvector equation.

Summary:

For $\theta = 90°$: Eigenvalues: $\lambda_1 = i, \lambda_2 = -i$ Eigenvectors: $\lambda_1 = i$: $\begin{bmatrix} 1 \\ -i \end{bmatrix}$ $\lambda_2 = -i$: $\begin{bmatrix} 1 \\ i \end{bmatrix}$

For $\theta = 180°$: Eigenvalue: $\lambda = -1$ (with algebraic multiplicity 2) Eigenvectors: Any non-zero vector in $\mathbb{R}^2$, for example, $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

Rubric:

- 1 point for correctly identifying each eigenvalue (times 2).
- 1 point for correctly identifying each eigenvector (times 4).
- 2 points for proper justification of answer (times 2).

c) Convert the given vectors ($\vec{v}_1, \vec{v}_2, \vec{v}_3,$ and $\vec{v}_4$) into orthonormal vectors using the Gram-Schmidt process. Verify your result by showing that the dot product of any two vectors in the obtained set is zero. 10 + 1 (CO3) (PO1)

Solution:

Given vectors are:

$$\vec{v}_1 = \begin{bmatrix} 2 \\ 0 \\ 2 \\ -1 \\ 0 \end{bmatrix}, \quad \vec{v}_2 = \begin{bmatrix} 0 \\ 3 \\ 0 \\ 0 \\ -4 \end{bmatrix}, \quad \vec{v}_3 = \begin{bmatrix} 1 \\ 1 \\ 2 \\ -3 \\ 2 \end{bmatrix}, \quad \vec{v}_4 = \begin{bmatrix} -1 \\ 2 \\ 1 \\ 3 \\ 1 \end{bmatrix}$$

We will apply the Gram-Schmidt process to these vectors to obtain an orthonormal basis.

Step 1: Orthonormalize $\vec{v}_1$

$$\vec{u}_1 = \vec{v}_1 = \begin{bmatrix} 2 \\ 0 \\ 2 \\ -1 \\ 0 \end{bmatrix}$$

$$||\vec{u}_1|| = \sqrt{2^2 + 0^2 + 2^2 + (-1)^2 + 0^2} = 3$$

$$\vec{e}_1 = \frac{\vec{u}_1}{||\vec{u}_1||} = \frac{1}{3}\begin{bmatrix} 2 \\ 0 \\ 2 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2/3 \\ 0 \\ 2/3 \\ -1/3 \\ 0 \end{bmatrix}$$

Step 2: Orthonormalize $\vec{v}_2$ (already perpendicular to $_1$)$\vec{u}_2 = \vec{v}_2 - \text{proj}_{\vec{u}_1}\vec{v}_2 = \vec{v}_2 - \frac{\vec{v}_2 \cdot \vec{u}_1}{\vec{u}_1 \cdot \vec{u}_1}\vec{u}_1 \vec{v}_2 \cdot \vec{u}_1 = (0)(2) + (3)(0) + (0)(2) + (0)(-1) + (-4)(0) = 0 Since_2 \cdot \vec{u}_1 = 0$, $\vec{v}_2$ is orthogonal to $\vec{u}_1$.

$$\vec{u}_2 = \vec{v}_2 = \begin{bmatrix} 0 \\ 3 \\ 0 \\ 0 \\ -4 \end{bmatrix}$$

$$||\vec{u}_2|| = \sqrt{0^2 + 3^2 + 0^2 + 0^2 + (-4)^2} = \sqrt{25} = 5$$

$$\vec{e}_2 = \frac{\vec{u}_2}{||\vec{u}_2||} = \frac{1}{5}\begin{bmatrix} 0 \\ 3 \\ 0 \\ 0 \\ -4 \end{bmatrix} = \begin{bmatrix} 0 \\ 3/5 \\ 0 \\ 0 \\ -4/5 \end{bmatrix}$$

Step 3: Orthonormalize $\vec{v}_3$

$$\vec{u}_3 = \vec{v}_3 - \text{proj}_{\vec{u}_1}\vec{v}_3 - \text{proj}_{\vec{u}_2}\vec{v}_3 = \vec{v}_3 - \frac{\vec{v}_3 \cdot \vec{u}_1}{\vec{u}_1 \cdot \vec{u}_1}\vec{u}_1 - \frac{\vec{v}_3 \cdot \vec{u}_2}{\vec{u}_2 \cdot \vec{u}_2}\vec{u}_2$$

$$\vec{v}_3 \cdot \vec{u}_1 = (1)(2) + (1)(0) + (2)(2) + (-3)(-1) + (2)(0) = 2 + 0 + 4 + 3 + 0 = 9$$

$$\vec{u}_1 \cdot \vec{u}_1 = 9$$

$$\vec{v}_3 \cdot \vec{u}_2 = (1)(0) + (1)(3) + (2)(0) + (-3)(0) + (2)(-4) = 0 + 3 + 0 + 0 - 8 = -5$$

$$\vec{u}_2 \cdot \vec{u}_2 = 25$$

$$\vec{u}_3 = \begin{bmatrix} 1 \\ 1 \\ 2 \\ -3 \\ 2 \end{bmatrix} - \frac{9}{9}\begin{bmatrix} 2 \\ 0 \\ 2 \\ -1 \\ 0 \end{bmatrix} - \frac{-5}{25}\begin{bmatrix} 0 \\ 3 \\ 0 \\ 0 \\ -4 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 2 \\ -3 \\ 2 \end{bmatrix} - \begin{bmatrix} 2 \\ 0 \\ 2 \\ -1 \\ 0 \end{bmatrix} + \frac{1}{5}\begin{bmatrix} 0 \\ 3 \\ 0 \\ 0 \\ -4 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \\ -2 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 3/5 \\ 0 \\ 0 \\ -4/5 \end{bmatrix} = \begin{bmatrix} -1 \\ 8/5 \\ 0 \\ -2 \\ 6/5 \end{bmatrix}$$

$$\|\vec{u}_3\| = \sqrt{(-1)^2 + (8/5)^2 + 0^2 + (-2)^2 + (6/5)^2} = \sqrt{9} = 3$$

$$\vec{e}_3 = \frac{\vec{u}_3}{\|\vec{u}_3\|} = \frac{1}{3}\begin{bmatrix} -1 \\ 8/5 \\ 0 \\ -2 \\ 6/5 \end{bmatrix} = \begin{bmatrix} -1/3 \\ 8/15 \\ 0 \\ -2/3 \\ 2/5 \end{bmatrix}$$

Step 4: Orthonormalize $\vec{v}_4$

$$\vec{u}_4 = \vec{v}_4 - \text{proj}_{\vec{u}_1}\vec{v}_4 - \text{proj}_{\vec{u}_2}\vec{v}_4 - \text{proj}_{\vec{u}_3}\vec{v}_4$$

$$\vec{u}_4 = \vec{v}_4 - \frac{\vec{v}_4 \cdot \vec{u}_1}{\vec{u}_1 \cdot \vec{u}_1}\vec{u}_1 - \frac{\vec{v}_4 \cdot \vec{u}_2}{\vec{u}_2 \cdot \vec{u}_2}\vec{u}_2 - \frac{\vec{v}_4 \cdot \vec{u}_3}{\vec{u}_3 \cdot \vec{u}_3}\vec{u}_3$$

$$\vec{v}_4 \cdot \vec{u}_1 = (-1)(2) + (2)(0) + (1)(2) + (3)(-1) + (1)(0) = -2 + 0 + 2 - 3 + 0 = -3$$

$$\vec{v}_4 \cdot \vec{u}_2 = (-1)(0) + (2)(3) + (1)(0) + (3)(0) + (1)(-4) = 0 + 6 + 0 + 0 - 4 = 2$$

$$\vec{v}_4 \cdot \vec{u}_3 = (-1)(-1) + (2)(8/5) + (1)(0) + (3)(-2) + (1)(6/5) = 1 + 16/5 + 0 - 6 + 6/5 = -5 + 22/5 = -3/5$$

$$\vec{u}_4 = \begin{bmatrix} -1 \\ 2 \\ 1 \\ 3 \\ 1 \end{bmatrix} - \frac{-3}{9}\begin{bmatrix} 2 \\ 0 \\ 2 \\ -1 \\ 0 \end{bmatrix} - \frac{2}{25}\begin{bmatrix} 0 \\ 3 \\ 0 \\ 0 \\ -4 \end{bmatrix} - \frac{-3/5}{9}\begin{bmatrix} -1 \\ 8/5 \\ 0 \\ -2 \\ 6/5 \end{bmatrix}$$

$$\vec{u}_4 = \begin{bmatrix} -1 \\ 2 \\ 1 \\ 3 \\ 1 \end{bmatrix} + \frac{1}{3}\begin{bmatrix} 2 \\ 0 \\ 2 \\ -1 \\ 0 \end{bmatrix} - \frac{2}{25}\begin{bmatrix} 0 \\ 3 \\ 0 \\ 0 \\ -4 \end{bmatrix} + \frac{1}{15}\begin{bmatrix} -1 \\ 8/5 \\ 0 \\ -2 \\ 6/5 \end{bmatrix} = \begin{bmatrix} -2/5 \\ 28/15 \\ 5/3 \\ 38/15 \\ 7/5 \end{bmatrix}$$

$$\|\vec{u}_4\| = \sqrt{(-2/5)^2 + (28/15)^2 + (5/3)^2 + (38/15)^2 + (7/5)^2} = \sqrt{74/5}$$

$$\vec{e}_4 = \frac{\vec{u}_4}{\|\vec{u}_4\|} = \frac{1}{\sqrt{74/5}}\begin{bmatrix} -2/5 \\ 28/15 \\ 5/3 \\ 38/15 \\ 7/5 \end{bmatrix} = \sqrt{\frac{5}{74}}\begin{bmatrix} -2/5 \\ 28/15 \\ 5/3 \\ 38/15 \\ 7/5 \end{bmatrix} = \begin{bmatrix} -2\sqrt{5}/(5\sqrt{74}) \\ 28\sqrt{5}/(15\sqrt{74}) \\ 5\sqrt{5}/(3\sqrt{74}) \\ 38\sqrt{5}/(15\sqrt{74}) \\ 7\sqrt{5}/(5\sqrt{74}) \end{bmatrix}$$

The orthonormal vectors are:

$$\vec{e}_1 = \begin{bmatrix} 2/3 \\ 0 \\ 2/3 \\ -1/3 \\ 0 \end{bmatrix}, \quad \vec{e}_2 = \begin{bmatrix} 0 \\ 3/5 \\ 0 \\ 0 \\ -4/5 \end{bmatrix}, \quad \vec{e}_3 = \begin{bmatrix} -1/3 \\ 8/15 \\ 0 \\ -2/3 \\ 2/5 \end{bmatrix}, \quad \vec{e}_4 = \begin{bmatrix} -2\sqrt{5}/(5\sqrt{74}) \\ 28\sqrt{5}/(15\sqrt{74}) \\ 5\sqrt{5}/(3\sqrt{74}) \\ 38\sqrt{5}/(15\sqrt{74}) \\ 7\sqrt{5}/(5\sqrt{74}) \end{bmatrix}$$

And, $\vec{e}_1 \perp \vec{e}_2 = 0$

Rubric:

- 1 point for correctly identifying first orthonormal vector.
- 2 point for correctly identifying second orthonormal vector.
- 3 point for correctly identifying third orthonormal vector.
- 4 point for correctly identifying fourth orthonormal vector.
- 1 point for showing dot product of any two orthonormal vectors.

3. The linear algebra course teacher of SWE 22, Professor Struggle, had been puzzled by the varying performance patterns in his class. He suspected there might be a direct relationship between how many hours students slept before their quizzes and the marks they achieved. To test his hypothesis, he meticulously tracked the sleep hours and corresponding quiz scores for five students over the semester. The results were compiled in the following Table 1.

Table 1: Table Representing the Correlation Between Sleep Duration and Quiz Scores of Students

Student	**Sleep Hours (x)**	**Quiz Marks (y)**
Mahbub	0	15
Awsaf	4	12
Mr. Strang	2	3
Pias	8	13
Utsho	16	9

a) An outlier is a data point that significantly deviates from the overall pattern in a dataset, often lying far from the regression line. Identify the outlier(based on intuition) among the five given points in Table 1 and briefly justify why it qualifies as an outlier. (Exclude the outlier from any calculations in the following questions.) 2 (CO3) (PO1)

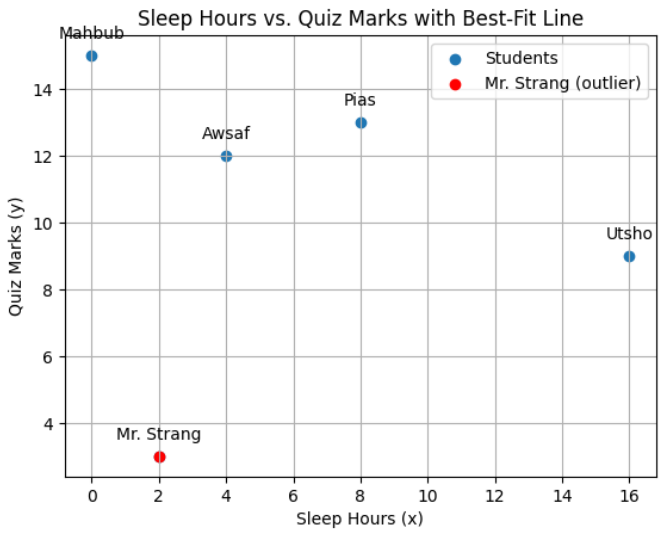

Figure 1: Quiz Marks vs Sleep Hours

Solution:

Plotting the points in a 2D graph as shown in Figure 1 clearly shows point representing Mr.Strang's info doesnt align with the pattern of the others.

Rubric:

- 1 point for correctly identifying outlier.
- 1 point for providing reason.

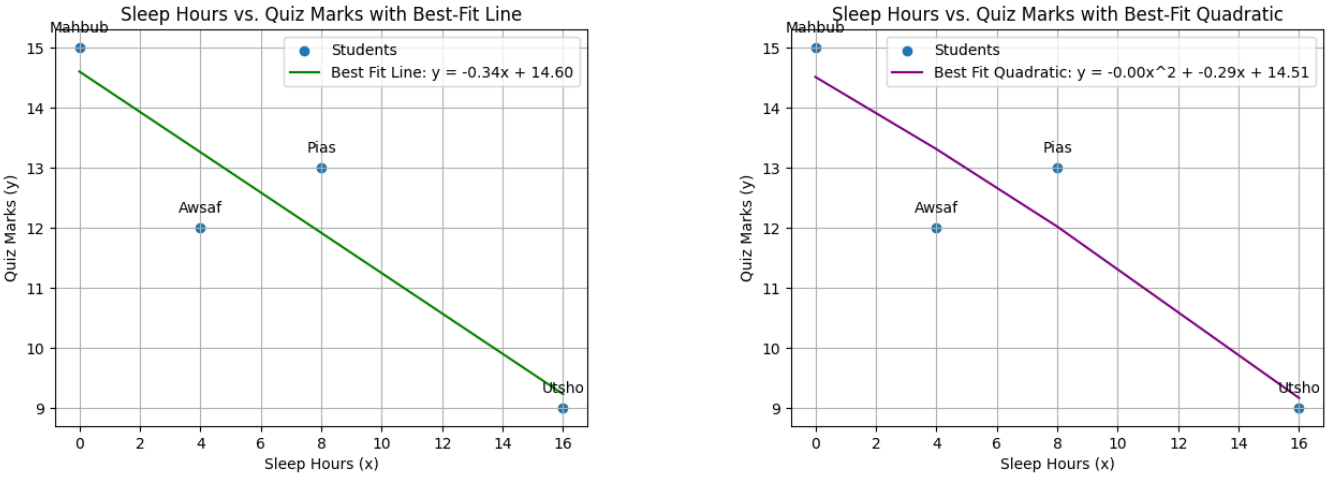

Figure 3: Best Fit Quadratic Curve

b) Using matrix algebra, determine the equation of the best-fit straight line, $y = mx + c$, for the four points (excluding the outlier) in Question 3. Also, find the error vector, e. 5 + 3 (CO3) (PO1)

Solution:

Let the equation of the best fit line be $y = mx + c$. The equations obtained by replacing the x and y from the points mentioned in Question 3 excluding the outlier are -

$$0 \times m + c = 15$$

$$4 \times m + c = 12$$

$$8 \times m + c = 13$$

$$13 \times m + c = 9$$

Representing them in matrix form we get,

$$Ax = b$$

$$\begin{bmatrix} 0 & 1 \\ 4 & 1 \\ 8 & 1 \\ 16 & 1 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} 15 \\ 12 \\ 13 \\ 9 \end{bmatrix}$$

$$\Rightarrow \tilde{x}, \begin{bmatrix} m \\ c \end{bmatrix} = (A^T \times A)^{-1} A^T \times b$$

$$\Rightarrow \tilde{x}, \begin{bmatrix} m \\ c \end{bmatrix} = (\begin{bmatrix} 0 & 4 & 8 & 16 \\ 1 & 1 & 1 & 1 \end{bmatrix} \times \begin{bmatrix} 0 & 1 \\ 4 & 1 \\ 8 & 1 \\ 16 & 1 \end{bmatrix})^{-1} \begin{bmatrix} 0 & 4 & 8 & 16 \\ 1 & 1 & 1 & 1 \end{bmatrix} \times \begin{bmatrix} 15 \\ 12 \\ 13 \\ 9 \end{bmatrix}$$

$$\Rightarrow \tilde{x} = \begin{bmatrix} \frac{-47}{140} \\ \frac{73}{5} \end{bmatrix} = \begin{bmatrix} -0.3357 \\ 14.6 \end{bmatrix}$$

Therefore, the best fit line is, $y = \frac{-47}{140}x + \frac{73}{5}$
Error vector,

$$e = b - p$$

$$\Rightarrow e = b - A\tilde{x}$$

$$\Rightarrow e = \begin{bmatrix} 15 \\ 12 \\ 13 \\ 9 \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ 4 & 1 \\ 8 & 1 \\ 16 & 1 \end{bmatrix} \begin{bmatrix} -0.3357 \\ 14.6 \end{bmatrix}$$

$$e = \begin{bmatrix} 0.4 \\ -1.257 \\ 1.08 \\ -0.2285 \end{bmatrix}$$

Rubric:

- 1 point for correctly identifying each matrix/vector (A and b) (times 2).
- 1 for correct expression to determine $\tilde{x}$.
- 2 points for correctly determining $\tilde{x}$.
- 1 point for correctly writing expression to determine e.
- 2 points for correctly determining e.

c) Determine the equation of the best-fit quadratic curve, $y = ax^2 + bx + c$, for the four points (excluding the outlier) in Question 3 using matrix algebra. Also, determine the residual vector (error vector), e, for this case. 5 + 3 (CO3) (PO1)

Solution:

Let the equation of the best fit line be $y = ax^2 + bx + c$. The equations obtained by replacing the x and y from the points mentioned in Question 3 excluding the outlier are -

$$0^2 \times a + 0 \times b + c = 15$$

$$4^2 \times a + 4 \times b + c = 12$$

$$8^2 \times a + 8 \times b + c = 13$$

$$16^2 \times a + 16 \times b + c = 9$$

Representing them in matrix form we get,

$$Ax = b$$

$$\begin{bmatrix} 0 & 0 & 1 \\ 16 & 4 & 1 \\ 64 & 8 & 1 \\ 256 & 16 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 15 \\ 12 \\ 13 \\ 9 \end{bmatrix}$$

$$\Rightarrow \tilde{x}, \begin{bmatrix} a \\ b \\ c \end{bmatrix} = (A^T \times A)^{-1} A^T \times b$$

$$\Rightarrow \tilde{x}, \begin{bmatrix} a \\ b \\ c \end{bmatrix} = (\begin{bmatrix} 0 & 0 & 1 \\ 16 & 4 & 1 \\ 64 & 8 & 1 \\ 256 & 16 & 1 \end{bmatrix}^T \times \begin{bmatrix} 0 & 0 & 1 \\ 16 & 4 & 1 \\ 64 & 8 & 1 \\ 256 & 16 & 1 \end{bmatrix})^{-1} \begin{bmatrix} 0 & 0 & 1 \\ 16 & 4 & 1 \\ 64 & 8 & 1 \\ 256 & 16 & 1 \end{bmatrix}^T \times \begin{bmatrix} 15 \\ 12 \\ 13 \\ 9 \end{bmatrix}$$

$$\Rightarrow \tilde{x} = \begin{bmatrix} \frac{-1}{352} \\ -\frac{127}{440} \\ \frac{798}{55} \end{bmatrix} = \begin{bmatrix} -0.0028 \\ -0.2886 \\ 14.51 \end{bmatrix}$$

Therefore, the best fit line is, $y = \frac{-1}{352}x^2 - \frac{127}{440}x + \frac{798}{55}$
Error vector,

$$e = b - p$$

$$\Rightarrow e = b - A\tilde{x}$$

$$\Rightarrow e = \begin{bmatrix} 15 \\ 12 \\ 13 \\ 9 \end{bmatrix} - \begin{bmatrix} 0 & 0 & 1 \\ 16 & 4 & 1 \\ 64 & 8 & 1 \\ 256 & 16 & 1 \end{bmatrix} \begin{bmatrix} \frac{-1}{352} \\ -\frac{127}{440} \\ \frac{798}{55} \end{bmatrix}$$

$$\Rightarrow e = \begin{bmatrix} 0.4909 \\ -1.309 \\ 0.9818 \\ -0.163 \end{bmatrix}$$

Rubric:

- 1 point for correctly identifying each matrix/vector (A and b) (times 2).
- 1 for correct expression to determine $\tilde{x}$.
- 2 points for correctly determining $\tilde{x}$.
- 1 point for correctly writing expression to determine e.
- 2 points for correctly determining e.

d) Which of the two representations obtained in Question 3.b and Question 3.c provides a better fit to the data and why? 4 (CO3) (PO1)

Solution:

For best fit line,

$$e = \begin{bmatrix} 0.4 \\ -1.257 \\ 1.08 \\ -0.2285 \end{bmatrix}$$

So, $E_1 = e_1^2 + e_2^2 + e_3^2 + e_4^2 = (0.4)^2 + (-1.257)^2 + (1.08)^2 + (-0.2285)^2 = 2.95886$
For best fit quadratic, curve,

$$e = \begin{bmatrix} 0.4909 \\ -1.309 \\ 0.9818 \\ -0.163 \end{bmatrix}$$

So, $E_1 = e_1^2 + e_2^2 + e_3^2 + e_4^2 = (0.4909)^2 + (-1.309)^2 + (0.9818)^2 + (-0.163)^2 = 2.94496$
Since $E_2 < E_1$, the quadratic equation obtained in Question 3.c provides better fit.

Rubric:

- 1.5 points for any logical comparison (times 2).
- 1 for correct justification.

e) If a new data point is introduced where a student who slept for 10 hours scored 10 marks on the quiz, which of the two equations obtained in Question 3.b and Question 3.c predicts the quiz score more accurately? Does this contradict your answer in Question 3.d? Justify. 3 (CO3) (PO1)

Solution:

Given point (10,10),
For eqn obtained in Question 3.b, $error_1 = |y_{actual} - y_{predicted}|$

$$\Rightarrow error_1 = |10 - (\frac{-47}{140}x + \frac{73}{5})|$$

Replacing $x = 10$,

$$\Rightarrow error_1 = 1.2429$$

For eqn obtained in Question 3.c, $error_2 = |y_{actual} - y_{predicted}|$

$$\Rightarrow error_1 = |10 - (\frac{-1}{352}x^2 - \frac{127}{440}x + \frac{798}{55})|$$

Replacing $x = 10$,

$$\Rightarrow error_2 = 1.344$$

$$error_2 > error_1$$

, this contradicts the previous statement, because the line is closer to the best fit line, however that doesn't mean the quadratic line is wrong, if there were more data points it would have given lower error.

Rubric:

- 2 points for any calculation for comparison.
- 1 for logical justification.

4. Given,

$$A = \begin{bmatrix} 1 & 3 & 2 \\ -2 & -6 & 1 \\ 2 & 5 & 7 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 4 & 5 & 6 & 4 & 5 & 6 \\ 7 & 8 & 9 & 7 & 8 & 9 \\ 1 & 2 & 3 & 1 & 2 & 3 \\ 4 & 5 & 6 & 4 & 5 & 6 \\ 7 & 8 & 9 & 7 & 8 & 9 \end{bmatrix}$$

a) Decompose the matrix A into LDU, where L, D and U are lower triangular, diagonal and upper triangular matrices, respectively. If necessary, multiply A by a permutation matrix P. 10 (CO2) (PO1)

Solution:

To decompose the matrix $A = \begin{bmatrix} 1 & 3 & 2 \\ -2 & -6 & 1 \\ 2 & 5 & 7 \end{bmatrix}$ into LDU, where L is a lower triangular matrix with ones on the diagonal, D is a diagonal matrix, and U is an upper triangular matrix with ones on the diagonal, we will perform Gaussian elimination.

Step 1: Eliminate the entry in the (2,1) position.

$R_2 = R_2 + 2R_1$

$$A^{(1)} = \begin{bmatrix} 1 & 3 & 2 \\ 0 & 0 & 5 \\ 2 & 5 & 7 \end{bmatrix}$$

The multiplier is $l_{21} = -2$.

Step 2: Eliminate the entry in the (3,1) position.

$R_3 = R_3 - 2R_1$

$$A^{(2)} = \begin{bmatrix} 1 & 3 & 2 \\ 0 & 0 & 5 \\ 0 & -1 & 3 \end{bmatrix}$$

The multiplier is $l_{31} = 2$.

Step 3: We encounter a zero in the (2,2) position. We need to perform a row swap to continue the LDU decomposition without a permutation matrix where L has ones on the diagonal.

Swap R_2 and R_3:

$$PA = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 3 & 2 \\ -2 & -6 & 1 \\ 2 & 5 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 5 & 7 \\ -2 & -6 & 1 \end{bmatrix} = A'$$

Now, perform Gaussian elimination on A'.

Step 1': Eliminate the entry in the (2,1) position of A'.

$R_2 = R_2 - 2R_1$

$$A'^{(1)} = \begin{bmatrix} 1 & 3 & 2 \\ 0 & -1 & 3 \\ -2 & -6 & 1 \end{bmatrix}$$

The multiplier is $l'_{21} = 2$.

Step 2': Eliminate the entry in the (3,1) position of $A'^{(1)}$.

$R_3 = R_3 + 2R_1$

$$A'^{(2)} = \begin{bmatrix} 1 & 3 & 2 \\ 0 & -1 & 3 \\ 0 & 0 & 5 \end{bmatrix} = U'$$

The multiplier is $l'_{31} = -2$.

Now we can form L' and D' from the multipliers and the diagonal elements of U'.

$$L' = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$$

$$D' = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$U' = \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix} \text{(Divide rows of } A'^{(2)} \text{ by the diagonal elements of } D'\text{)}$$

So, $A' = L'D'U'$.

$$\begin{bmatrix} 1 & 3 & 2 \\ 2 & 5 & 7 \\ -2 & -6 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix}$$

Since $PA = A'$, we have $A = P^{-1}A' = P^TA' = P^TL'D'U'$.

$$P^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$L = P^TL' = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix}$$

$$D = D' = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$U = U' = \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix}$$

So, $A = LDU$:

$$\begin{bmatrix} 1 & 3 & 2 \\ -2 & -6 & 1 \\ 2 & 5 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\boxed{A = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix}}$$

Rubric:

- 2 point for correctly performing each elimination (times 3).
- 1 point for considering permutation.
- 2 point for correctly separating D and U.
- 1 points each for correct answer.

b) Compute B^2 using the properties of matrix multiplication. 7 (CO2) (PO1)

Solution:

Given,

$$B = \begin{bmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 4 & 5 & 6 & 4 & 5 & 6 \\ 7 & 8 & 9 & 7 & 8 & 9 \\ 1 & 2 & 3 & 1 & 2 & 3 \\ 4 & 5 & 6 & 4 & 5 & 6 \\ 7 & 8 & 9 & 7 & 8 & 9 \end{bmatrix}$$

This can't be solved using a regular calculator but if you observe carefully at the question, there is some repetition,
Lets define

$$X = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Then, B can be represented using X,

$$B = \begin{bmatrix} X & X \\ X & X \end{bmatrix}$$

$$\therefore B^2 = \begin{bmatrix} X & X \\ X & X \end{bmatrix} \begin{bmatrix} X & X \\ X & X \end{bmatrix}$$

$$\Rightarrow B^2 = \begin{bmatrix} X^2 + X^2 & X^2 + X^2 \\ X^2 + X^2 & X^2 + X^2 \end{bmatrix}$$

$$\Rightarrow B^2 = \begin{bmatrix} 2X^2 & 2X^2 \\ 2X^2 & 2X^2 \end{bmatrix}$$

Now,

$$X^2 = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$X^2 = \begin{bmatrix} 30 & 36 & 42 \\ 66 & 81 & 96 \\ 102 & 126 & 150 \end{bmatrix}$$

Rubric:

- 1 point for identifying pattern.
- 2 points for correctly calculating X^2.
- 2 points for correctly determining that each segment will be $2X^2$.
- 2 points for correct answer.

c) Show that the eigenvalues of a projection matrix P are $\lambda = 0$ and $\lambda = 1$. Explain the reasoning behind finding these eigenvalues and what they represent. 6 + 2 (CO3) (PO1)

Solution:

Let P be a projection matrix.

$\Rightarrow\ Px = \lambda x$

$\Rightarrow\ P^2x = \lambda x \quad$ [For projection matrix, $P = P^2$]

$\Rightarrow\ P(Px) = \lambda x$

$\Rightarrow\ P(\lambda x) = \lambda x \quad [Px = \lambda x]$

$\Rightarrow\ \lambda(Px) = \lambda x$

$\Rightarrow\ \lambda(\lambda x) = \lambda x$

$\Rightarrow\ \lambda^2 x = \lambda x$

$\Rightarrow\ \lambda^2 x - \lambda x = 0$

$\Rightarrow\ (\lambda^2 - \lambda)x = 0$

Since $x \neq$ zero vector, we have:

$\lambda^2 - \lambda = 0$

$\lambda(\lambda - 1) = 0$

$\Rightarrow\ \lambda = 0, 1$

Let the projection matrix of $\vec{a} = \begin{bmatrix} a \\ b \end{bmatrix}$ be P.

$\Rightarrow\ P = \frac{\vec{a}\vec{a}^T}{\vec{a}^T\vec{a}} = \frac{1}{a^2+b^2}\begin{bmatrix} a \\ b \end{bmatrix}\begin{bmatrix} a & b \end{bmatrix} = \frac{1}{a^2+b^2}\begin{bmatrix} a^2 & ab \\ ab & b^2 \end{bmatrix}$ For $\lambda = 1$,

$$\begin{aligned}
(P - \lambda I) &= \frac{1}{a^2+b^2}\begin{bmatrix} a^2 & ab \\ ab & b^2 \end{bmatrix} - 1\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\
&= \frac{1}{a^2+b^2}\begin{bmatrix} a^2 & ab \\ ab & b^2 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\
&= \frac{1}{a^2+b^2}\begin{bmatrix} a^2 - (a^2+b^2) & ab \\ ab & b^2 - (a^2+b^2) \end{bmatrix} \\
&= \frac{1}{a^2+b^2}\begin{bmatrix} -b^2 & ab \\ ab & -a^2 \end{bmatrix}
\end{aligned}$$

$\therefore$ Nullspace of the given vector, eigenvector $= \begin{bmatrix} a \\ b \end{bmatrix}$

For $\lambda = 0$, $(P - \lambda I) = \frac{1}{a^2+b^2}\begin{bmatrix} a^2 & ab \\ ab & b^2 \end{bmatrix}$

$\therefore$ Eigenvector $= \begin{bmatrix} -b \\ a \end{bmatrix} \quad (\perp \begin{bmatrix} b \\ -a \end{bmatrix})$

Takeaways

1. Eigenvalues for Projection matrix is always either 0 or 1.
2. Eigenvector when $\lambda = 1$, is the vector whose responsible for projection matrix P.
3. Eigenvector when $\lambda = 0$, is the vector perpendicular to the vector for which we determined P.

$$x_1 \cdot x_2 = 0$$

Rubric:

- 1 point each for correctly identifying eigenvalues (times 2).
- 2 points each for determining eigenvectors or any valid argument. (times 2)
- 1 point each for correct justification (times 2).

5. a) The Network Flow Equilibrium Law states that for each router in a network, the sum of incoming packet flows must equal the sum of outgoing packet flows. Consider a network with four routers and six packet flows: packet 1 (p_1), packet 2 (p_2), ..., packet 6 (p_6) illustrated in Figure 4, where arrows show the direction of flow. For example, at Router 1, the flow equation is:

$$p_1 + p_4 = p_3$$

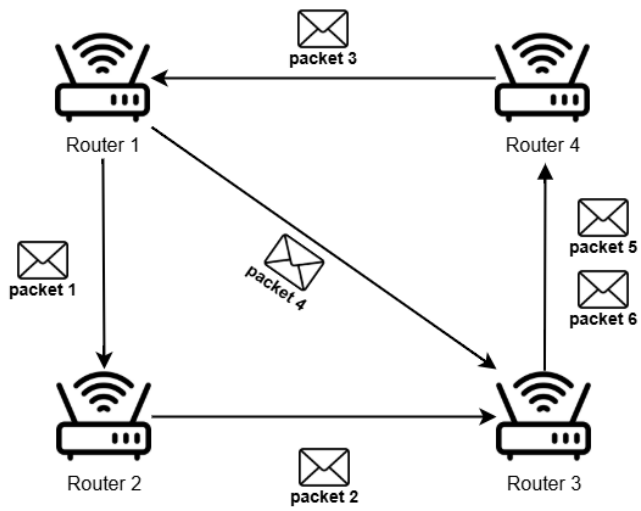

Figure 4: Packet flow between routers in a network for Question 5.a.

i. Express the network in Figure 4 using a matrix. 1 (CO2) (PO1)

Table 2: Network Flow Equilibrium Equations

	p_1	p_2	p_3	p_4	p_5	p_6
Router 1	-1	0	+1	-1	0	0
Router 2	+1	-1	0	0	0	0
Router 3	0	+1	0	+1	-1	-1
Router 4	0	0	-1	0	+1	+1

Solution:

The Network Flow Equilibrium Law states that for each router in a network, the sum of incoming packet flows must equal the sum of outgoing packet flows. Considering the network in Figure 4 with four routers and six packet flows ($p_1, p_2, p_3, p_4, p_5, p_6$), the flow equations at each router are:

- **Router 1:** $p_1 + p_4 = p_3 \Rightarrow -p_1 + p_3 - p_4 = 0$
- **Router 2:** $p_2 = p_1 \Rightarrow +p_1 - p_2 = 0$
- **Router 3:** $p_2 + p_4 = p_5 + p_6 \Rightarrow p_2 + p_4 + p_5 + p_6 = 0$
- **Router 4:** $p_3 = p_5 + p_6 \Rightarrow -p_3 + p_5 + p_6 = 0$

Rubric:

- 1 point each for correctly identifying Matrix.

ii. Convert the matrix obtained in Question 5.a)i into Row Reduced Echelon Form (RREF). 5 (CO2) (PO1)

Solution:

Find the reduced row echelon form of
Given:

$$\begin{bmatrix} -1 & 0 & 1 & -1 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & -1 & -1 \\ 0 & 0 & -1 & 0 & 1 & 1 \end{bmatrix}$$

Step 1: Get a leading 1 in the first position. Multiply row 1 by -1: $R_1 \leftarrow -1 \times R_1$

$$\begin{bmatrix} 1 & 0 & -1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & -1 & -1 \\ 0 & 0 & -1 & 0 & 1 & 1 \end{bmatrix}$$

Step 2: Eliminate the 1 in position (2,1). Subtract row 1 from row 2: $R_2 \leftarrow R_2 - R_1$

$$\begin{bmatrix} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & -1 & 0 & 0 \\ 0 & 1 & 0 & 1 & -1 & -1 \\ 0 & 0 & -1 & 0 & 1 & 1 \end{bmatrix}$$

Step 3: Get a 1 in position (2,2). Multiply row 2 by -1: $R_2 \leftarrow -1 \times R_2$

$$\begin{bmatrix} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & -1 & -1 \\ 0 & 0 & -1 & 0 & 1 & 1 \end{bmatrix}$$

Step 4: Eliminate the 1 in position (3,2). Subtract row 2 from row 3: $R_3 \leftarrow R_3 - R_2$

$$\begin{bmatrix} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 & -1 \\ 0 & 0 & -1 & 0 & 1 & 1 \end{bmatrix}$$

Step 5: Get a 1 in position (4,3). Multiply row 4 by -1: $R_4 \leftarrow -1 \times R_4$

$$\begin{bmatrix} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 & -1 \\ 0 & 0 & 1 & 0 & -1 & -1 \end{bmatrix}$$

Step 6: Eliminate the duplicate row. Since rows 3 and 4 are identical, we can keep just one of them.

$$\begin{bmatrix} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Step 7: Eliminate all other entries in column 3. Add row 3 to row 1: $R_1 \leftarrow R_1 + R_3$

$$\begin{bmatrix} 1 & 0 & 0 & 1 & -1 & -1 \\ 0 & 1 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Add row 3 to row 2: $R_2 \leftarrow R_2 + R_3$

$$\begin{bmatrix} 1 & 0 & 0 & 1 & -1 & -1 \\ 0 & 1 & 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Therefore, the reduced row echelon form is:

$$\begin{bmatrix} 1 & 0 & 0 & 1 & -1 & -1 \\ 0 & 1 & 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Rubric:

- 5 point for correct RREF form.

iii. Find all possible combinations of packet flows $(p, p, ..., p)$ for which the Conservation of Data Packets principle is satisfied at every router. In other words, find the complete solution determining its nullspace and particular solution. 4 (CO2) (PO1)

Solution:

Nullspace:

$$\begin{bmatrix} -1 \\ -1 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Since the Right hand sides of the equations are 0, this is the solution.

Rubric:

- 1 point for each correct nullspace (times 3).
- 1 point for pointing it out that there are no particular solutions.

b) Given,

$$N = \begin{bmatrix} 1 & 3 & 2 & 1 & 4 \\ 2 & 6 & 1 & 3 & 5 \\ 3 & 9 & 0 & 2 & 6 \end{bmatrix}$$

i. Find out the Four Fundamental Subspaces of matrix N (rowspace, columnspace, nullspace and left nullspace). For each vectorspace mention its Dimension and bases. 4×3 (CO2) (PO1)

Solution:

$$Given, \begin{bmatrix} 1 & 3 & 2 & 1 & 4 \\ 2 & 6 & 1 & 3 & 5 \\ 3 & 9 & 0 & 2 & 6 \end{bmatrix}$$

RREF

Subtract row 1 multiplied by 2 from row 2: $R_2 \leftarrow R_2 - 2R_1$.

$$\begin{bmatrix} 1 & 3 & 2 & 1 & 4 \\ 0 & 0 & -3 & 1 & -3 \\ 3 & 9 & 0 & 2 & 6 \end{bmatrix}$$

Subtract row 1 multiplied by 3 from row 3: $R_3 \leftarrow R_3 - 3R_1$.

$$\begin{bmatrix} 1 & 3 & 2 & 1 & 4 \\ 0 & 0 & -3 & 1 & -3 \\ 0 & 0 & -6 & -1 & -6 \end{bmatrix}$$

Since the element at row 2 and column 2 (pivot element) equals 0, we need to swap the rows. Find the first nonzero element in column 2 under the pivot entry. As can be seen, there are no such entries. Move to the next column.

Divide row 2 by -3: $R_2 \leftarrow -\frac{1}{3}R_2$.

$$\begin{bmatrix} 1 & 3 & 2 & 1 & 4 \\ 0 & 0 & 1 & -\frac{1}{3} & 1 \\ 0 & 0 & -6 & -1 & -6 \end{bmatrix}$$

Subtract row 2 multiplied by 2 from row 1: $R_1 \leftarrow R_1 - 2R_2$.

$$\begin{bmatrix} 1 & 3 & 0 & \frac{5}{3} & 2 \\ 0 & 0 & 1 & -\frac{1}{3} & 1 \\ 0 & 0 & -6 & -1 & -6 \end{bmatrix}$$

Add row 2 multiplied by 6 to row 3: $R_3 \leftarrow R_3 + 6R_2$.

$$\begin{bmatrix} 1 & 3 & 0 & \frac{5}{3} & 2 \\ 0 & 0 & 1 & -\frac{1}{3} & 1 \\ 0 & 0 & 0 & -3 & 0 \end{bmatrix}$$

Divide row 3 by -3: $R_3 \leftarrow -\frac{1}{3}R_3$.

$$\begin{bmatrix} 1 & 3 & 0 & \frac{5}{3} & 2 \\ 0 & 0 & 1 & -\frac{1}{3} & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Subtract row 3 multiplied by $\frac{5}{3}$ from row 1: $R_1 \leftarrow R_1 - \frac{5}{3}R_3$.

$$\begin{bmatrix} 1 & 3 & 0 & 0 & 2 \\ 0 & 0 & 1 & -\frac{1}{3} & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Add row 3 multiplied by $\frac{1}{3}$ to row 2: $R_2 \leftarrow R_2 + \frac{1}{3}R_3$.

$$\begin{bmatrix} 1 & 3 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

ANSWER

The reduced row echelon form is

$$\begin{bmatrix} 1 & 3 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Columnspace :

Base =3

Bases = $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}$

Nullspace :

Base =2

Bases = $\begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}$

To find the Row Reduced Echelon Form (RREF) of the transpose of matrix N, where

$$N = \begin{bmatrix} 1 & 3 & 2 & 1 & 4 \\ 2 & 6 & 1 & 3 & 5 \\ 3 & 9 & 0 & 2 & 6 \end{bmatrix}$$

The transpose of N is:

$$N^T = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 6 & 9 \\ 2 & 1 & 0 \\ 1 & 3 & 2 \\ 4 & 5 & 6 \end{bmatrix}$$

Now, we perform row reduction on N^T:

$$\begin{bmatrix} 1 & 2 & 3 \\ 3 & 6 & 9 \\ 2 & 1 & 0 \\ 1 & 3 & 2 \\ 4 & 5 & 6 \end{bmatrix} \xrightarrow{R_2-3R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 3 & 2 \\ 4 & 5 & 6 \end{bmatrix} \xrightarrow{R_3-2R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & -3 & -6 \\ 1 & 3 & 2 \\ 4 & 5 & 6 \end{bmatrix} \xrightarrow{R_4-R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & -3 & -6 \\ 0 & 1 & -1 \\ 4 & 5 & 6 \end{bmatrix}$$

$$\xrightarrow{R_5-4R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & -3 & -6 \\ 0 & 1 & -1 \\ 0 & -3 & -6 \end{bmatrix} \xrightarrow{SwapR_2 \leftrightarrow R_4} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -1 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \\ 0 & -3 & -6 \end{bmatrix} \xrightarrow{R_3+3R_2} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -1 \\ 0 & 0 & -9 \\ 0 & 0 & 0 \\ 0 & -3 & -6 \end{bmatrix}$$

$$\xrightarrow{R_5+3R_2} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -1 \\ 0 & 0 & -9 \\ 0 & 0 & 0 \\ 0 & 0 & -9 \end{bmatrix} \xrightarrow{-\frac{1}{9}R_3} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & -9 \end{bmatrix} \xrightarrow{R_5+9R_3} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow{R_1-3R_3} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_2+R_3} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1-2R_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The Row Reduced Echelon Form (RREF) of N^T is:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Rowspace :
Base =3
Bases = $\begin{bmatrix} 1 \\ 3 \\ 2 \\ 1 \\ 4 \end{bmatrix}, \begin{bmatrix} 2 \\ 6 \\ 1 \\ 3 \\ 5 \end{bmatrix}, \begin{bmatrix} 3 \\ 9 \\ 0 \\ 2 \\ 6 \end{bmatrix}$
Nullspace :
Base =0
Bases = Nullvector

Rubric:

- 1 point each for correctly identifying number of bases (times 4).
- 2 points each for correct identification of the bases (times 4).

ii. Determine which pairs of subspaces obtained in Question 5.b)i are orthogonal. Justify your answer by showing that the dot product of any basis vector from one subspace with any basis vector from its orthogonal counterpart is zero. 3 (CO2) (PO1)

Solution:

Columnspace $\perp$ Left Nullspace
Rowspace $\perp$ Nullspace

Rubric:

- 1.5 points each for each orthogonal pair (times 2).

6. Given $Ax = b$, where

$$A = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix}, \; x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \; b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

where,

$$det(A_1) = 8 \, , \; det(A_2) = -11 \, , \; det(A_3) = 3$$

$$and \; C = \begin{bmatrix} 2 & 0 & 0 & 7 \\ 3 & 0 & 4 & 0 \\ 0 & 5 & -1 & 0 \\ 0 & -2 & 0 & 3 \end{bmatrix}$$

A_i is the matrix similar to matrix A but its $i-$th column is replaced by vector b, for example

$$A_1 = \begin{bmatrix} b_1 & 2 & 1 \\ b_2 & 1 & -1 \\ b_3 & 1 & 1 \end{bmatrix}$$

a) Retrieve vector x and b using your knowledge of Cramer's rule and the provided information in Question 6. 10 (CO2) (PO1)

Solution:

Given the system of linear equations $Ax = b$, where:

$$A = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

We are also given the determinants of matrices A_1, A_2, and A_3, which are formed by replacing the i-th column of A with the vector b:

$$\det(A_1) = 8$$
$$\det(A_2) = -11$$
$$\det(A_3) = 3$$

To use Cramer's rule, we first need to find the determinant of matrix A:

$$\det(A) = 3 \cdot \det \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} - 2 \cdot \det \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} + 1 \cdot \det \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$
$$\det(A) = 3 \cdot (1 \cdot 1 - (-1) \cdot 1) - 2 \cdot (2 \cdot 1 - (-1) \cdot 1) + 1 \cdot (2 \cdot 1 - 1 \cdot 1)$$
$$\det(A) = 3 \cdot (1 + 1) - 2 \cdot (2 + 1) + 1 \cdot (2 - 1)$$
$$\det(A) = 3 \cdot 2 - 2 \cdot 3 + 1 \cdot 1$$
$$\det(A) = 6 - 6 + 1$$
$$\det(A) = 1$$

Now, we can use Cramer's rule to find the components of the vector x:

$$x_1 = \frac{\det(A_1)}{\det(A)} = \frac{8}{1} = 8$$
$$x_2 = \frac{\det(A_2)}{\det(A)} = \frac{-11}{1} = -11$$
$$x_3 = \frac{\det(A_3)}{\det(A)} = \frac{3}{1} = 3$$

So, the vector x is:

$$x = \begin{bmatrix} 8 \\ -11 \\ 3 \end{bmatrix}$$

Next, we need to find the vector b. We know that $Ax = b$, so we can multiply matrix A by the vector x we just found:

$$b = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 8 \\ -11 \\ 3 \end{bmatrix}$$

$$b = \begin{bmatrix} 3(8) + 2(-11) + 1(3) \\ 2(8) + 1(-11) + (-1)(3) \\ 1(8) + 1(-11) + 1(3) \end{bmatrix}$$

$$b = \begin{bmatrix} 24 - 22 + 3 \\ 16 - 11 - 3 \\ 8 - 11 + 3 \end{bmatrix}$$

$$b = \begin{bmatrix} 5 \\ 2 \\ 0 \end{bmatrix}$$

Rubric:

- 2 point for correctly determining $det(A)$.
- 3 points for correctly determining x.
- 3 points for correctly determining b.
- 1 points each for correct answer (times 2).

b) Apply the False Expansion Theorem on matrix A to show that - 7 (CO2) (PO1)

$$\sum_{i=1}^{n} a_{xi} C_{yi} = 0 \quad (\text{where } x \neq y)$$

While solving clearly mention each a_{xi} and C_{yi} that you require. (x and y represents row number, n represents total number of rows/columns, a_{ij} presents the entry along the i-th column and j-th row of matrix A and C_{ij} represents the co-factor of entry a_{ij}.)

Solution:

Lets take $x = 1$ and $y = 2$,
so,

$$\sum_{i=1}^{3} a_{1i} C_{2i} = a_{11} C_{21} + a_{12} C_{22} + a_{13} C_{23}$$

$$= -3 \begin{vmatrix} 2 & 1 \\ 1 & 1 \end{vmatrix} + 2 \begin{vmatrix} 3 & 1 \\ 1 & 1 \end{vmatrix} - 1 \begin{vmatrix} 3 & 2 \\ 1 & 1 \end{vmatrix}$$

$$= -3(2-1) + 2(3-1) - 1(3-2)$$

$$= 0$$

$$[Showed]$$

Rubric:

- 1 points for correctly identifying each co-factor with sign (times 3).
- 3 points for correct calculation.

- 1 point for correctly showing it equals to 0.

c) Determine the determinant of matrix C using The Big Formula. 8 (CO2) (PO1)

Solution:

Given,

$$C = \begin{bmatrix} 2 & 0 & 0 & 7 \\ 3 & 0 & 4 & 0 \\ 0 & 5 & -1 & 0 \\ 0 & -2 & 0 & 3 \end{bmatrix}$$

Applying the Big Formula,

$$det(C) = \begin{vmatrix} 2 & 0 & 0 & 0 \\ 3 & 0 & 4 & 0 \\ 0 & 5 & -1 & 0 \\ 0 & -2 & 0 & 3 \end{vmatrix} + \begin{vmatrix} 0 & 0 & 0 & 7 \\ 3 & 0 & 4 & 0 \\ 0 & 5 & -1 & 0 \\ 0 & -2 & 0 & 3 \end{vmatrix}$$

$$det(C) = \begin{vmatrix} \boxed{2} & 0 & 0 & 0 \\ 3 & 0 & \boxed{4} & 0 \\ 0 & \boxed{5} & -1 & 0 \\ 0 & -2 & 0 & \boxed{3} \end{vmatrix} + \begin{vmatrix} 0 & 0 & 0 & \boxed{7} \\ \boxed{3} & 0 & 4 & 0 \\ 0 & 5 & \boxed{-1} & 0 \\ 0 & \boxed{-2} & 0 & 3 \end{vmatrix}$$

$$det(C) = -(2 \times 4 \times 5 \times 3) + (7 \times 3 \times -2 \times -1) = -78$$

(The first part requires odd number of exchanges and the second part requires even number of exchanges.)

Rubric:

- 2 point for correct decomposition.
- 5 points for correct process.
- 1 point for correct answer.